Angles - Mark scheme

Geometry · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10

Question 1

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · November 2025 · Question 9

Q	Working	Answer	Mark	Notes
(i)		35	1	B1 allow seen on diagram if no contradiction
(ii)		110	1	B1 ft their value of x, ie $180 - 2x$ allow seen on diagram if no contradiction
				Total 2 marks

Question 2

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2021 · Question 11

Q	Working	Answer	Mark	Notes
	[interior angle of pentagon =] 540 ÷ 5 (= 108) oe or [exterior angle of pentagon =] 360 ÷ 5 (= 72)		3	M1 for a correct calculation for an interior or an exterior angle of a regular pentagon
	360 – (90 + "108") or 90 + "72" or 180 – ("108" – 90) oe			M1 for a fully correct method "108" or "72" must come from correct working and be used correctly
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	162		A1
				Total 3 marks

Question 3

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2019 · Question 2

2	Ext ∠ = 180 – 162 (= 18) oe **or** $\frac{(n-2)180}{n} = 162$ oe		3	M1
	360 ÷ "18" oe **or** $18n = 360$			M1
		20		A1
				Total 3 marks

Question 4

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2023 · Question 4

4	3 × 180 (= 540) or 360 – [(180 – 90) + (180 – 135) + (180 – 67) + (180 – 119)] (= 51) or 360 – (90 + 45 + 113 + 61) (= 51)		3	M1
	90 + 135 + 67 + 119 + x = "540" oe 411 + x = "540" oe or "540" – (90 + 135 + 67 + 119) or 3 × 180 – (90 + 135 + 67 + 119) oe or 540 – 411 or 180 – "51" oe			M1
	Correct answer scores full marks (unless from obvious incorrect working)	129		A1
				Total 3 marks

Angles - Mark scheme

Question 5

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2026 · Question 6

6	$180 \times (5-2)$ or $5 \times \left(180 - \frac{360}{5}\right)$ $(= 540)$ oe **OR** $(180-96)+(180-91)+(180-93)+2y = 360$ or $84+89+87+2y = 360$ **or** $(180-96)+(180-91)+(180-93)$ $(= 84+89+87 = 260)$ or $360-(180-96)-(180-91)-(180-93)$ $(= 100)$		3	M1	for a correct method to find the sum of the interior angles **OR** forming an equation involving exterior angles (y is the exterior angle at E and B) may use any letter **or** finding the total of the exterior angles for the known angles or for the unknown angles
	eg $(2x =)$ "540" $- 96 - 91 - 93$ $(=$ "540" $- 280 = 260)$ or $(x =) \frac{"540"-96-91-93}{2}$ oe $\left(= \frac{"540"-280}{2}\right)$ **OR** $(y =) \frac{"100"}{2}$ $(=50)$ or $(x =) 180 - \frac{"100"}{2}$ oe			M1	(dep on M1) for getting as far as the value for $2x$ **or** for a complete method for x using interior angles **OR** for finding the exterior angle at E or B **or** for a complete method using exterior angles
	Correct answer scores full marks (unless from obvious incorrect working).	130		A1	cao
					Total 3 marks

Question 6

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2026 · Question 15

Q	Working	Answer	Mark	Notes
	eg $(5-2) \times 180$ $(= 540)$ **or** $(10-2) \times 180$ $(= 1440)$ **or** $(10-2) \times 180 \div 10$ $(= 144)$ **or** $360 \div 10$ $(= 36)$		3	M1 for a method to find the sum of the interior angles for a pentagon or a decagon **or** the size of an interior angle of a regular decagon **or** the size of an exterior angle of a regular decagon Angles may be seen on the diagram
	eg "540" $-$ (80 + 75 + 109 + 180 + "36") or "540" $-$ (80 + 75 + 109 + 360 $-$ "144") **or** 80 + 75 + 109 + 180 + "36" + x = "540"			M1 for a method to the missing angle in the pentagon **or** for forming an equation in x
	Correct answer scores full marks (unless from obvious incorrect working)	60		A1
				Total 3 marks

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Angles - Mark scheme

Question 7

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2026 · Question 7

Question		Working	Answer	Mark	Notes
*	(a)	$360-85-60-90$ **or** $85+60+90 = 235$ **and** $360 - "235"$		2	M1
		Correct answer scores full marks (unless from obvious incorrect working)	125		A1
	(b)	$7a+5a=180$ **or** $12a=180$ **or** $(a=)\frac{180}{12}$ **or** $(a=)\frac{180}{7+5}$ **or** $(7a=)105$ **or** $(5a=)75$		2	M1
		Correct answer scores full mark (unless from obvious incorrect working)	15		A1
					Total 4 marks

Question 8

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2026 · Question 7

Q	Working	Answer	Mark	Notes
	$BAC = 52$ **or** EAC / $DAB = 180 - 52$ $(= 128)$		4	M1 for $BAC = 52$ or method to find angle EAC or DAB Angles to be identified either by notation or correctly positioned on the diagram
	$(BCA) = 360 - 329$ $(= 31)$			M1 for a method to find angle BCA
	$(ABC =)$ $180 - ("52" + "31")$ **or** $180 -$ [their BAC] $-$ [their BCA] **or** "128" – "31" **or** [their EAC / DAB] – [their BCA]			M1 for use of angles in a triangle = 180 [their BAC] is their found angle for BAC [their BCA] is their found angle for BCA **or** for use of exterior angle is equal to sum of two opposite interior angles [their EAC / DAB] is their found angle for EAC or DAB
	Correct answer scores full marks (unless from obvious incorrect working)	97		A1 cao
				Total 4 marks

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Question 9

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · January 2023 · Question 9

Q	Working	Answer	Mark	Notes
	(*ABD* =) 360 – 52 – 112 – 90 (= 106)		4	M1 may be marked in correct place on diagram
	(*CBD* =) 180 – "106" (=74)			M1 may be marked in correct place on diagram
		32		A1
		Reasons given		B1 dep on M1 At least two appropriate reasons given. "angles in a quadrilateral add to 360°" accept 4-sided shape. "angles on a straight line add to 180°" **or** angles on a straight line add to 180° "angles in a triangle add to 180°" **or** angles in a triangle sum to 180° "base angles in an isosceles triangle (are equal)"
				Total 4 marks

Question 10

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2026 · Question 15

Question	Working	Answer	Mark	Notes	
	360 – 314 (= 46)		4	M1 angles may be marked on the diagram	M2 for $\frac{314-180}{2}$
	180 – "46" (= 134)			M1 do not award this mark for 314 – 180 (=134)	
	Correct angle scores 3 marks (unless from obvious incorrect working)	67		A1	
				B1 dep on M2 for two correct reasons from (i) angles around a point add up to 360° (ii) angles in a triangle add to 180° **or** angles in a triangle add to 180° (iii) base angle in an isosceles triangle (are equal) **OR** dep on M2 for The exterior angle of a triangle is equal to the sum of the interior opposite angles	
					Total 4 marks

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Question 11

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2023 · Question 3

3	$(4x - 27) + (3x + 46) = 180$ oe or "expression for C" + $(3x + 10) = 180$ **or** $7x + 19 = 180$ **or** $3x + 46 + 4x - 27 + 3x + 10 + [\text{"}180 - (3x + 10)\text{"}] = 360$		4	M1	Sum angles A and B to 180, or find an expression for BCD and sum all angles to 360. [condone missing brackets and condone use of any letter or expression for angle C (even x or BCD)]
				A1	$x = 23$
	eg $3 \times$"23" + 46 (= 115) **or** eg $180 - (3 \times$"23" + 10) (= 101)			M1ft	dep on M1 using **their** x to calculate a value for angle B or 'their' C (cannot be a negative value and cannot just be x)
	Correct answer scores full marks (unless from obvious incorrect working)	115		A1	Allow $3x + 46$ or ABC if 115 is clearly seen in working or on diagram
					Total 4 marks

Question 12

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2022 · Question 4

4	$360 \div 8$ (= 45) or $360 \div 5$ (= 72) or $180 - (360 \div 8)$ (= 135) oe or $180 - (360 \div 5)$ (= 108) oe		4	M1 finding interior or exterior angle of octagon or pentagon Angles may be seen on diagram – but must be obtuse if interior and acute if exterior.
	'72' – '45' (= 27) or '135' – '108' (= 27)			M1 (dep 1st M1) using a pair of interior or pair of exterior angles to find angle IBC Angle may be seen on diagram.
	$\frac{180 - '27'}{2}$ (= 76.5)			M1
		76.5		A1
				Total 4 marks

Question 13

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2024 · Question 5

5	$(5 - 2) \times 180$ (= 540) **or** $360 \div 5$ (= 72)		4	M1 NB If angles are on the diagram they must be from correct working and correctly assigned
	$\frac{"540"}{5}$ (= 108) or $180 -$ "72" (= 108) **or** $180 - 96$ (= 84)			M1
	"72" + "84" **or** $360 - (96 +$ "108") **or** $180 -$ ("108" – "84")			M1 for a complete method
	Working required	156		A1 dep on M2
				Total 4 marks

Question 14

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · January 2019 · Question 6

6	$3x + 10 = x + 52$	34	4	M1	for equating the expressions for angle P and angle Q
	$3x - x = 52 - 10$ **or** $2x = 42$ **or** $x = 21$			M1	for isolating the terms in x
	$y = 180 - 2 \times (\text{"}21\text{"} + 52)$ **or** $y = 180 - 2 \times (3 \times \text{"}21\text{"} + 10)$ **or** $y = 180 - (\text{"}21\text{"} + 52) - (3 \times \text{"}21\text{"} + 10)$			M1	for a complete method
				A1	dep on M2

Angles - Mark scheme

Question 15

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2022 · Question 19

Q	Working	Answer	Mark	Notes	
	$7x + 3x + 8x = 360$ oe		4	M1	M2 for $7x = 140$ (140 can be on diagram)
	$(x =)\ 360 \div 18\ (= 20)$			M1	
	$360 \div (180 - 7 \times$ "20") oe or $360 \div (180 -$ "140") $\frac{(n-2)\times 180}{n} = 7 \times$ "20" oe or $360 \div 40$			M1 for 360 ÷ exterior angle	
		9		A1	
					Total 4 marks

Question 16

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2025 · Question 20

Q	Working		Answer	Mark	Notes
	eg $360 - (148 + 50)\ (= 162)$ **or** $180 - 50\ (= 130)$ **or** $180 - 148\ (= 32)$			4	M1 for method to find the interior angle of the polygon **or** start to the method of finding the exterior angle of the polygon
	eg $180 -$ "162" (= 18) **or** $148 -$ "130" (= 18) **or** $50 -$ "32" (= 18)	eg $180(n-2) =$ "162" n **or** $180(n-2) \div n =$ "162"			M1 for method to find the exterior angle **or** for setting up an equation using sum of interior angles formula
	eg $360 \div$ "18"	eg $(n =)\ 360 \div (180 -$ "162")			M1 for a complete method
	Working required		20		A1 dep on M1
					Total 4 marks

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Question 17

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · November 2020 · Question 21

Q	Working	Answer	Mark	Notes
	$(10-2)\times 180$ oe (= 1440) **or** $(6-2)\times 180$ oe (= 720)		4	M1 for a method to find the sum of the interior angles of a decagon or a hexagon
	'1440' – 148 – 2×150 – 2×168 – 2×134 – 2×125 (=138) **or** '1440' – 1302 (= 138) **or** '720' – 148÷2 – 150 – 168 – 134 – 125 (= 69) **or** '720' – 651 (= 69)			M1 Allow omission of one angle
	360 – '138' **or** 360 – 2 × '69'			M1
		222		A1
	Alternative method (exterior angles)			
	360 – 2×(180 – 125) – 2×(180 – 134) – 2×(180 – 168) – 2×(180 – 150) – (180 – 148) **or** 360 – 2×55 – 2×46 – 2×12 – 2×30 – 32		4	M2 If not M2 then award M1 for at least 3 or (180 – 125) , (180 – 134) , (180 – 168) , (180 – 150), (180 – 148) **or** at least 3 of 55, 46, 12, 30, 32
	180 + '42'			M1
		222		A1
				Total 4 marks

Question 18

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2023 · Question 11

11	$SCD = 128°$ **or** $BCS = 32°$ **or** $TSC = 180 - 128$ (= 52)		4	M1	angles need to be identified or may be seen marked on the diagram	M2 for (*BCD* =) 128 + 32 (= 160) **or** (*DCV* =) 52 – 32 (= 20) (may be seen marked on the diagram). To award these marks 160 or 20 must be clearly used or identified as the interior or exterior angle.
	eg (int ∠ =)128 + 32(= 160) **or** (ext ∠ =)180 – (128 + 32)(= 20) **or** (ext ∠ =)"52" – 32(= 20)			M1	(dep on previous M1) for method to find the size of one interior or exterior angle, may be seen marked on the diagram.	
	eg $180(n-2) =$ "160"n **or** 360 ÷ "20"			M1	for setting up an equation for the sum of interior angles **or** 360 ÷ "20"	
	Working required	18		A1	dep on M2	
						Total 4 marks

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Angles - Mark scheme

Question 19

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2025 · Question 7

Q	Working	Answer	Mark	Notes
(a)		Reason given	1	B1 vertically opposite angles are equal **or** vertically opposite angles are equal opposite to 58(°)
(b)	(JGH =) 360 – 163 – 90 – 47 (= 60) **or** (JGH =) 180 – 110 (= 70)		4	M1 for a method to find angle JGH either using the quadrilateral **or** assuming line FGH is straight
	Correct answer scores full marks (unless from obvious incorrect working)			A1 (110 + 60 =) 170 **or** 110 + 60 ≠ 180 **or** (180 – 60 =) 120 **or** (70 + 163 + 90 + 47 =) 370 **or** 70+ 163 + 90 + 47 ≠ 360 **or** for (180 – 110 =) 70 **and** (360 – 163 – 90 – 47 =) 60
				B1 angles on a straight line add to 180° **or** angles on a straight line add to 180°
				B1 angles in a quad(rilateral) add up to 360 **or** angles in a quad(rilateral) add up to 360 (Accept 4-sided shape)
				Total 5 marks

Question 20

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · January 2022 · Question 8

Q	Working	Answer	M	Notes
(a)	(DBC =) 180 – (93 + 42) (= 45) **OR** (x =) 93 + 42		2	M1 for method to find angle DBC **OR** using exterior angle is equal to the sum of the two opposite interior angles
		135		A1
(b)(i)	360 – (90 + 100 + 114) oe		2	M1 for a complete method to find y
		56		A1
(ii)		Angles at a point sum to 360°	1	B1
				Total 5 marks

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CHALLENGE QUESTIONS

Question 21 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2024 · Question 12*

Q	Working	Answer	Mark	Notes
	$CED = 180 - 125\ (= 55)$ or $BCD = 360 - (86 + 78 + 128)\ (= 68)$		5	M1 angles may be marked on diagram or labelled unambiguously in working
	$DCE = 180 - (2 \times$ "55"$)(= 70)$			M1 angles may be marked on diagram or labelled unambiguously in working
	Correct answer of 138 scores 3 marks (unless from obvious incorrect working)	138		A1 for 138
				B2 (dep on M2) for all correct reasons for their method (B1 (dep on M1) for one correct reason for their method) Angles on a straight line add to 180 Angles on a straight line add to 180 Isosceles triangle Angles in a triangle add to 180 Angles in a triangle add to 180 Angles in a quadrilateral add to 360 Angles in a quadrilateral add to 360
				Total 5 marks

Question 22 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2021 · Question 5*

5	$(5-2) \times 180 - 112 - 102 - 96\ (= 230)$ oe eg $540 - 112 - 102 - 96\ (= 230)$ **or** $360 - (180 - 112) - (180 - 102) - (180 - 96)$ $(= 360 - 68 - 78 - 104 = 360 - 230 = 130)$ oe		5	M1
	$\frac{'540' - 112 - 102 - 96}{2}\ (= 115)$ **or** '130' ÷ 2 (= 65)			M1 dep on previous mark
	$\frac{180 \times (8-2)}{8}\ (= 135)$ **or** $180 - (360 \div 8)\ (= 135)$ **or** $\frac{360}{8}\ (= 45)$ as exterior angle of octagon			M1 indep Withhold the mark for $\frac{360}{8}\ (= 45)$ if shown as an interior angle
	360 – '115' – '135' **or** '65' + '45'			M1
	Working required	110		A1 dep on M1
				Total 5 marks

Angles - Mark scheme

Question 23 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2021 · Question 5*

Question	Working	Answer	Mark	Notes
5	Angle *EBC* or *ECB* = (180 – 44) ÷ 2 (= 68)		5	M1 Could be seen on diagram
	Angle *GBC* = 180 – "68" (= 112) **or** Angle *GBC* = "68" + 44 (= 112) **or** Angle *BGH* = "68" (same as *EBC*) Angle *ABE* = 180 – "68" (= 112) **and** Angle *BGF* = "112" **or** Angle *ABG* = "68" **and** Angle *BGH* = "68" or Angle *FGJ* = "68" **or** Angle *BGF* = 180 – "68"(=112)			M1 for a method to as far as one step away from working out Angle *JGH* (an angle corresponding or vertically opposite to *JGH* or at the same point on a straight line with *JGH*) Could be seen on diagram. (the award of this mark also implies the previous M1)
	Working not required, so correct angle scores 3 marks (unless from obvious incorrect working)	112		A1 Could be seen in correct place on diagram
	NB: reasons must include the underlined words *Accept ∠ for angle(s) and △ for triangle* **For all angles**: They must be clearly stated as the correct angle or shown on the diagram in the correct position. (eg just seeing 68 in working without a label is not sufficient for the award of a mark for angle *EBC*)			B2 for correct answer with full reasons for their method eg isosceles triangle (or 2 equal sides, 2 equal angles) Angles in a triangle sum to 180° or angles in a triangle Angles on a straight line sum to 180° Angles on a straight line sum to 180° Exterior angle in a triangle is equal to the two opposite interior angles. Vertically opposite angles are equal. Vertically opposite angles are equal. Corresponding angles are equal. Alternate angles are equal Allied angles sum to 180° (or co-interior angles) Angles at a point (or full turn) add up to 360° (or angles at a point) (B1 for one correct reason appropriate to their method, dep on M1)
				Total 5 marks

Question 24 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · January 2021 · Question 8*

8	(5 – 2) × 180 ÷ 5 (= 108) **or** 360 ÷ 5 (= 72)		5	M1	for method to find an interior or exterior angle of a pentagon
	(6 – 2) × 180 ÷ 6 (= 120) **or** 360 ÷ 6 (= 60)			M1	for method to find an interior or exterior angle of a hexagon
	360 – 108 – 120 (= 132) **or** 60 + 72 (= 132) **or** (180 – '120') + (180 – '108')			M1	dep on M2 for a correct method to find angle *EDI* using correct figures
	360 – '72' – '60' – '132' (= 96)			M1	for a complete method to find angle *x*
		96		A1	dep on correct working
				Note:	Angles may be seen on diagram throughout
					Total 5 marks

Question 25 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2024 · Question 11*

11	int angle of pentagon = (3 × 180) ÷ 5 (= 108) oe **or** ext angle of pentagon = 360 ÷ 5 (= 72) oe		5	M1	allow in working but not if labelled in wrong place on diagram (unless clearly started again)
	int angle of hexagon = (4 × 180) ÷ 6 (= 120) **or** ext angle of hexagon = 360 ÷ 6 (= 60)			M1	allow in working but not if labelled in wrong place on diagram (unless clearly started again)
	360 – ("108" + "120") (= 132) oe **or** "60" + "72" (= 132) **or** $(180-"108")+(180-"120")$ (= 132)			M1	A fully correct method to find the size of obtuse angle *AEF* but not if labelled in wrong place on diagram [Figures in inverted commas must come from correct working]
	$[180-("60"+"72")]\div 2$ oe **or** $\frac{180-"132"}{2}$ oe **or** $[180-(180-"108")-(180-"120")]\div 2$ oe			M1	A fully correct method to find the size of angle *EAF* [Figures in inverted commas must come from correct working]
	Correct answer scores full marks (unless from obvious incorrect working)	24		A1	cao
					Total 5 marks